Contents

1. Introduction

Agriculture and poultry farming play a vital role in the economy of many developing countries. Modern poultry farms require continuous monitoring of feeding systems, water availability, and security. Traditional poultry farm management relies heavily on manual labour, which is time-consuming, increases operational cost, and is prone to human error.

This project presents an **IoT-Based Smart Poultry Farm Monitoring and Automation System** built around a dual-controller architecture using an **Arduino Uno** and a **Raspberry Pi**. The system automates gate control, feed dispensing, and water-level monitoring through low-cost sensors and actuators. The Arduino Uno performs real-time sensing and control, while the Raspberry Pi acts as the monitoring and data-processing unit. The result is a low-cost solution that improves farm-management efficiency, reduces labour requirements, and provides a foundation for future IoT-based smart-farming applications such as cloud dashboards and mobile alerts.

Keywords: Smart Poultry Farm, Arduino Uno, Raspberry Pi, IoT, Automation, Ultrasonic Sensor, Water Sensor, Servo Motor, Smart Agriculture.

2. Objectives

- To automate poultry **feeding** operations.
- To monitor **water levels** continuously and provide alerts when the level becomes low.
- To automate **gate** opening and closing.
- To collect and monitor farm data using a **Raspberry Pi**.
- To provide a low-cost, expandable platform that can later support cloud and mobile integration.

3. Problem Statement

Manual labour: Traditional poultry farm management depends on continuous human supervision for feeding, watering, and security. This raises operational cost, is time-consuming, and is prone to errors when a worker is unavailable or distracted.

Feeding & water: Irregular feeding and unnoticed shortages of drinking water directly affect the health and productivity of the flock. A farmer cannot watch the feed container and water trough every minute of the day.

Security & access: Farm gates are often operated manually, which is inconvenient and offers no automatic record of activity. An automated, sensor-driven system can address these gaps by performing routine tasks reliably and notifying the farmer only when attention is required.

4. Scope

- **Automatic gate control:** ultrasonic detection of an approaching object opens the gate via a servo motor and closes it after a fixed delay.
- **Automatic feeding:** an ultrasonic sensor measures the feed level and triggers a servo motor to dispense feed when the level drops below a threshold.
- **Water monitoring:** a water sensor continuously checks water availability and raises a buzzer alarm on shortage.

- **Data monitoring:** the Arduino transmits sensor values to the Raspberry Pi over serial communication for processing, logging, and dashboard display.
- **Future expansion:** the architecture is designed to add cloud monitoring, a mobile app, and additional environmental sensors.

5. Literature Review

Several researchers have developed smart poultry-farming systems using IoT and embedded technologies.

Study 1

A self-powered IoT-based smart poultry-farm system was developed to monitor environmental parameters and automate farm operations. The study demonstrated improved efficiency and reduced energy consumption **[1]**.

Study 2

An IoT-based poultry-farm monitoring system was proposed using real-time data-analysis techniques. The system monitored feed levels, water levels, and environmental conditions remotely **[2]**.

Study 3

A smart poultry-management system using Arduino-based automation was developed to automate feeding and environmental monitoring, reducing human intervention **[3]**, **[5]**.

Other studies emphasise farm security and safe, authorised access control **[4]**. Collectively, these studies indicate that automation and IoT technologies significantly improve poultry-farm productivity and management efficiency.

6. Methodology

The project follows an iterative cycle that integrates the hardware modules into a simple rule engine running on the Arduino, with the Raspberry Pi providing higher-level monitoring. The Arduino reads the ultrasonic and water sensors, applies threshold-based decisions, and drives the servo motors and buzzer. Sensor values are then streamed to the Raspberry Pi over serial communication for logging and display.

6.1 System Architecture

The proposed system consists of two processing units:

1. Arduino Uno — handles low-level, real-time operations: reading the ultrasonic sensors, reading the water-sensor data, controlling the servo motors (gate and feed), and activating the buzzer alarm.

2. Raspberry Pi — works as the monitoring and high-level data-processing unit: receiving sensor data through serial communication, processing and logging it, and enabling future cloud and mobile integration.

3. Sensor Layer: Gate ultrasonic sensor (HC-SR04), Feed ultrasonic sensor (HC-SR04), and a Water Sensor Module.

4. Actuation Layer: two servo motors (gate and feed dispenser) and a buzzer for the low-water alarm, powered from a stable external 5 V supply.

5. Application Layer: a monitoring dashboard on the Raspberry Pi for real-time data viewing, event logging, and future remote access.

System Flow: Sensors → Arduino Uno → Raspberry Pi → Monitoring Dashboard

6.2 Technology Stack

- **Hardware:** Arduino Uno, Raspberry Pi, HC-SR04 ultrasonic sensors, water sensor module, servo motors.
- **Firmware/Software:** Arduino (C/C++) for the controller; Python for the Raspberry Pi monitoring services.
- **Communication:** USB serial communication between the Arduino and the Raspberry Pi.
- **Power Management:** a regulated external 5 V supply with a common ground shared between the Arduino, servos, and Raspberry Pi.

7. Hardware Components

Category	Component	Qty	Purpose
Control	Arduino Uno	1	Real-time sensing and actuator control.
Control	Raspberry Pi	1	Monitoring, data processing, and logging.
Gate	HC-SR04 Ultrasonic Sensor	1	Detect an approaching object at the gate.
Feeding	HC-SR04 Ultrasonic Sensor	1	Measure feed level in the container.
Drive	Servo Motor	2	Open/close the gate and dispense feed.
Water	Water Sensor Module	1	Detect water availability / shortage.
Alarm	Buzzer	1	Audible low-water alert for the farmer.
Prototyping	Breadboard & Jumper Wires	—	Circuit assembly and interconnection.
Power	External 5 V Power Supply	1	Stable supply for servos and modules.

8. Budget Estimation

The following table summarises the estimated hardware budget based on the component list.

Item Description	Unit Price (BDT)	Qty	Total (BDT)
Raspberry Pi 3 (4GB)	14,100	1	14,100
MicroSD Card (32 GB, Class 10)	1,000	1	1,000
Arduino Uno	427	1	427
HC-SR04 Ultrasonic Sensor	90	2	180
Servo Motor (SG90 / MG-series)	120	2	240

Item Description	Unit Price (BDT)	Qty	Total (BDT)
Water Level Sensor Module	49	1	49
Passive Buzzer	15	1	15
Breadboard	120	1	120
Jumper Wires (set)	200	1	200
External 5 V Power Supply (2A)	300	1	300
Grand Total (Estimated)	**—**	**—**	**16,631**

* Prices are indicative local-market estimates (BDT) for budgeting purposes and may vary by vendor.

9. Work Plan (Timeline)

The work plan is organised into design, implementation, integration, and validation phases. Parallel development is used where possible (e.g., firmware and dashboard).

Phase / Task	Month 1	Month 2	Month 3	Month 4
Requirements & component selection	✔			
Software / hardware setup	✔			
Automatic gate control (ultrasonic + servo)	✔	✔		
Automatic feeding system		✔		
Water monitoring + buzzer alarm		✔	✔	
Serial communication & data logging			✔	
Monitoring dashboard			✔	✔
Integration testing & documentation				✔

10. Working Principle

1. Sensors collect real-time information (gate proximity, feed level, water level).
2. The Arduino processes the sensor readings.
3. Based on threshold values, control decisions are made.
4. Servo motors perform gate and feed automation.
5. A water shortage activates the buzzer alarm.
6. The Raspberry Pi receives the operational data over serial communication.
7. Data is stored and visualised for monitoring purposes.

Automatic gate control: an ultrasonic sensor is placed near the poultry gate; when an object is detected within a predefined distance, the Arduino drives a servo motor to open the gate, and the gate closes automatically after a fixed delay.

Automatic feeding: a second ultrasonic sensor monitors the feed level; when it drops below the threshold, a servo motor rotates to release additional feed into the container.

Water monitoring: a water sensor continuously monitors availability; if the level becomes insufficient, the buzzer generates an alarm to notify the farmer.

11. Results and Discussion

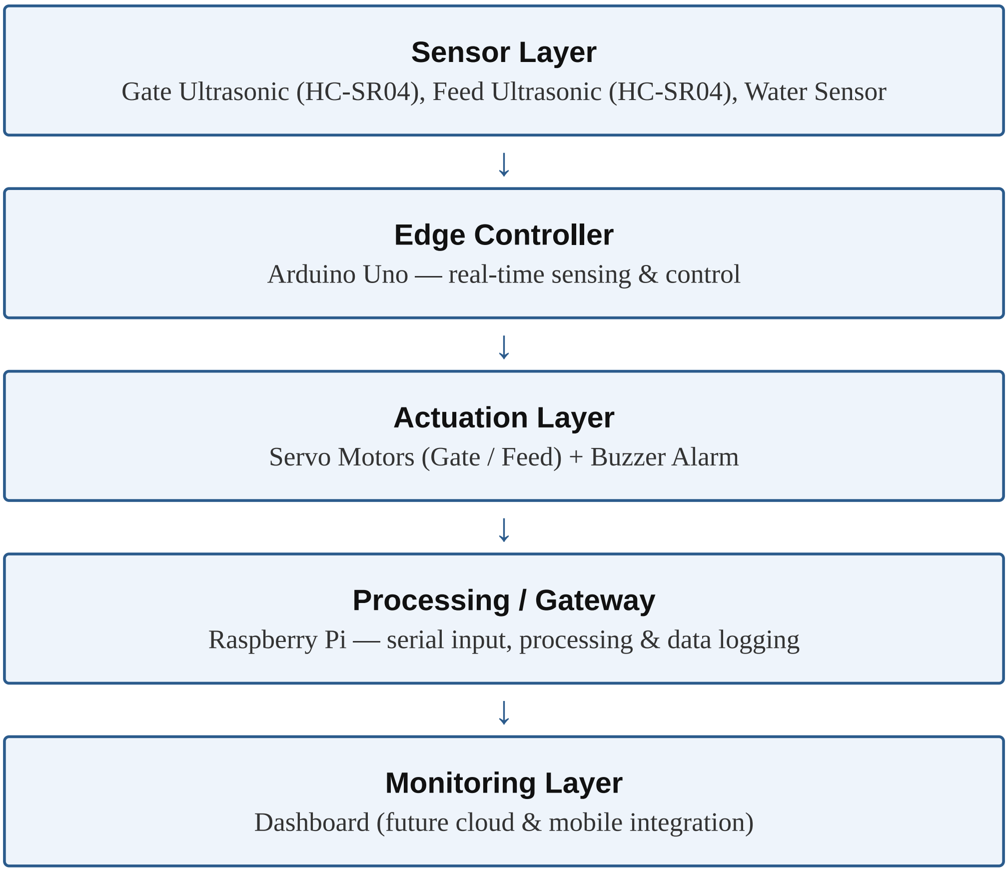
The developed prototype successfully demonstrated:

- Automatic gate control.
- Automatic feed dispensing.
- Water-shortage detection.
- Alarm generation.
- Real-time monitoring support through the Raspberry Pi.

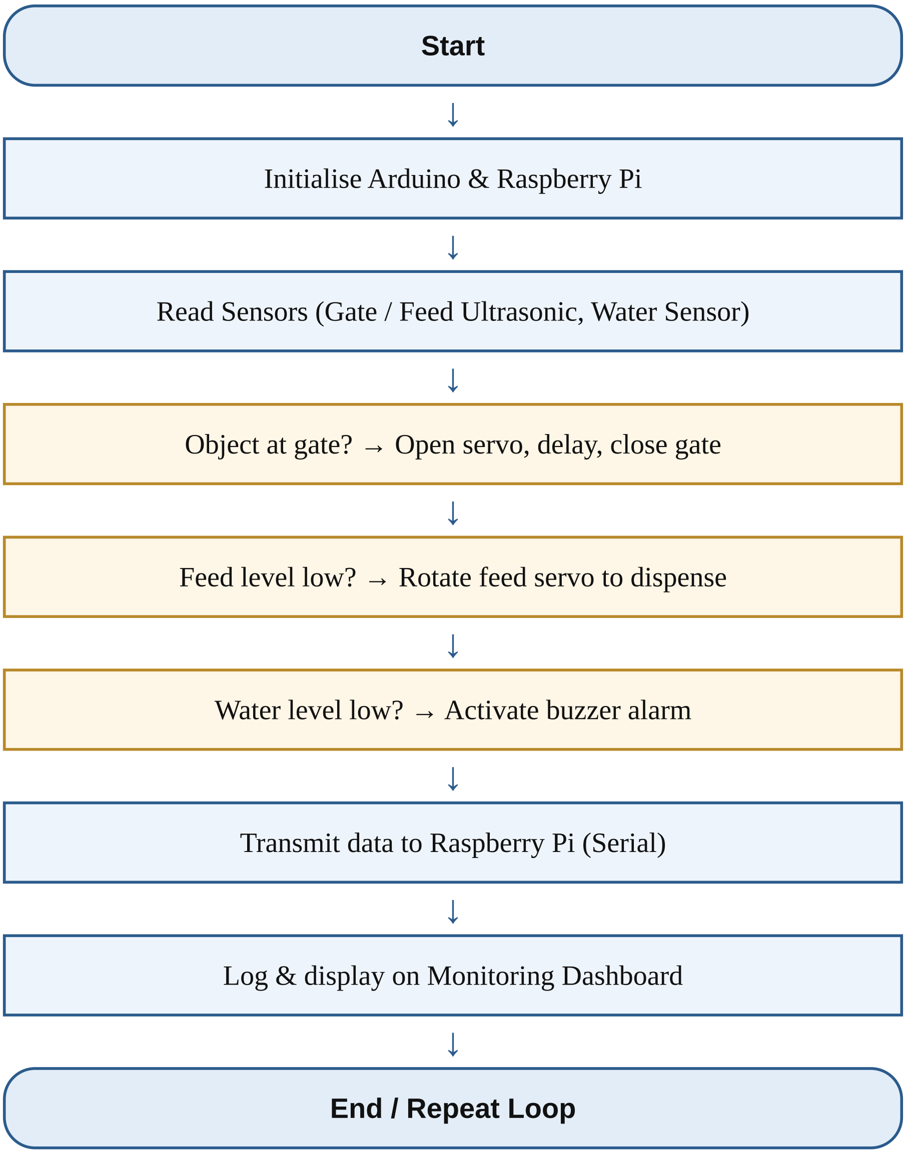
The system reduced manual intervention and showed stable operation during testing.

12. Visual Models

12.1 Block Diagram



12.2 Flow Chart



13. Advantages

- Low-cost implementation.
- Reduced labour requirements.
- Improved resource management.
- Real-time monitoring capability.
- Expandable IoT architecture.
- Easy maintenance and operation.

14. Limitations

- No internet connectivity in the current prototype.
- Sensor accuracy depends on environmental conditions.
- Requires a stable power supply.
- Limited remote-access functionality.

15. Future Scope

The system can be enhanced by integrating:

- Bluetooth monitoring and mobile-application control.
- A cloud-based dashboard for remote access.
- Temperature and humidity sensors for environmental control.
- A Raspberry Pi Camera Module for visual surveillance.
- AI-based poultry health monitoring and automatic disease detection.
- Smart security systems.

16. Conclusion

This project successfully demonstrates the implementation of an IoT-Based Smart Poultry Farm Monitoring and Automation System using Arduino Uno and Raspberry Pi. The system automates feeding, gate control, and water monitoring while providing a platform for future IoT integration. The dual-controller architecture keeps the system responsive to real-time events while supporting higher-level data processing on the Raspberry Pi. The proposed solution reduces labour cost, improves productivity, and supports the development of modern smart-farming technologies.

17. References

[1] “Self-Powered IoT-Based Design for Multi-Purpose Smart Poultry Farm,” Springer Nature. Available: https://doi.org/10.1007/978-981-15-7062-9_5

[2] “Implementation of IoT-Based Poultry Farm Monitoring System with Real-Time Data Analysis,” IEEE Xplore. Available: https://ieeexplore.ieee.org/document/11160018

[3] “IoT-Based Smart Poultry Management System,” ResearchGate. Available: https://www.researchgate.net/publication/380119628

[4] “IoT-Based Smart System for Safe and Secure Poultry Farming.” International Journal on Recent and Innovation Trends in Computing and Communication. Available: https://ijritcc.org/index.php/ijritcc/article/view/11141

[5] “Development of Smart Chicken Poultry Farm Using RTOS on Arduino,” IEEE Xplore. Available: https://ieeexplore.ieee.org/document/9057310